POWER: USB 5V (D101) / LiPo (Q101) auto power path -> VSYS -> U6 LP5907 3.0 V -> +3V0_A (ADS1298 AVDD)
U6 EN = XIAO 3V3, so the AFE is off whenever the ESP32 is off. DVDD = XIAO 3V3.

XIAO ESP32-S3
U2
ADS_RESET 1
ADS_START 2
ADS_CS 4
ADS_DRDY 5
UART_TX 7
D0_IO1
D1_IO2
D2_IO3
D3_IO4
D4_IO5
D5_IO6
TX_IO43
BAT+
BAT-
VBAT 15
AGND 16
5V 14 VBUS
GND 13 AGND
3V3 12 +3V3
D10_IO9 11 SPI_MOSI
D9_IO8 10 SPI_MISO
D8_IO7 9 SPI_SCK
RX_IO44 8 UART_RX

B5819W
D101
VBUS
VSYS
#FLG07
PWR_FLAG
VSYS
AO3401A
Q101
R104 100K
VBUS
AGND
VBUS 1
VBAT 3
VSYS 2
C107 10uF
VSYS
AGND

LP5907MFX-3.0
U6
VSYS 1
+3V3 3
IN
EN
OUT 5 +3V0_A
GND
AGND 2
C101 1uF +3V0_A AGND
C102 100nF +3V0_A AGND

TEST POINTS
TP1 +3V0_A
TP2 +3V3
TP3 AGND
TP4 VSYS
TP5 RLD_OUT
TP6 ADS_DRDY
TP7 SPI_SCK
TP8 AGND

LiPo (protected)
J3
1 VBAT
2 AGND
#FLG04
PWR_FLAG
VBAT
#FLG02
PWR_FLAG
AGND
LiPo: J3 wires to the XIAO underside BAT+/BAT- pads (U2 pins 15/16). XIAO is hand-soldered.

C120 100nF +3V0_A AGND
C119 100nF +3V0_A AGND
C118 10uF +3V0_A AGND
C121 1uF +3V3 AGND
C122 100nF +3V3 AGND
RLD: 1M || 1.5nF feedback, 330k series to the REF electrode(s) limits DC fault current to <10 uA.

DEBUG: USB serial over the XIAO USB-C is the default.
U5/J2 isolated UART is optional and DNP. Populate U5, C701, C702, J2 if needed;
J2 side is powered by the USB-UART adapter (3.3 or 5 V).

INPUT RC: 22k series (limits fault and plug-insertion current) + 1nF COG to AGND, ~7 kHz.
ADS1298 oversamples at 512 kHz, so this is the only anti-alias filter needed.

CH1 (TRS)
J1
S RLD_ELEC
R ELEC1_N
T ELEC1_P
ELEC1_P R201 2.2k CH1_P
ELEC1_N R202 2.2k CH1_N
CH1_P C201 1nF COG AGND
CH1_N C202 1nF COG AGND

CH2 (TRS)
J4
S RLD_ELEC
R ELEC2_N
T ELEC2_P
ELEC2_P R203 2.2k CH2_P
ELEC2_N R204 2.2k CH2_N
CH2_P C203 1nF COG AGND
CH2_N C204 1nF COG AGND

CH3 (TRS)
J5
S RLD_ELEC
R ELEC3_N
T ELEC3_P
ELEC3_P R205 2.2k CH3_P
ELEC3_N R206 2.2k CH3_N
CH3_P C205 1nF COG AGND
CH3_N C206 1nF COG AGND

ADS1298IPAGR
U1
+3V0_A 19 AVDD
+3V0_A 54 AVDD1
+3V3 48 DVDD
CH1_P 16 IN1P
CH1_N 15 IN1N
CH2_P 14 IN2P
CH2_N 13 IN2N
CH3_P 12 IN3P
CH3_N 11 IN3N
+3V0_A 10 IN4P
+3V0_A 9 IN4N
+3V0_A 8 IN5P
+3V0_A 7 IN5N
+3V0_A 6 IN6P
+3V0_A 5 IN6N
+3V0_A 4 IN7P
+3V0_A 3 IN7N
+3V0_A 2 IN8P
+3V0_A 1 IN8N
VCAP1 28
VCAP2 30
VCAP3 55
VCAP4 26
VREFP 24
AGND 25 VREFN
WCT 64
RLDOUT 63 RLD_OUT
RLDIN 62 RLD_ELEC
RLDINV 61 RLD_INV
RLDREF 60
TESTP_PACE_OUT1 17 +3V0_A
TESTP_PACE_OUT2 18 +3V0_A
GPIO1 42
GPIO2 44
GPIO3 45
GPIO4 46
DAISY_IN 41 AGND
SCLK 40 SPI_SCK
DIN 34 SPI_MOSI
CS 39 ADS_CS
DOUT 43 SPI_MISO
CLK 37
CLKSEL 52 +3V3
START 38 ADS_START
DRDY 47 ADS_DRDY
PWDN 35 ADS_PWDN
RESET 36 ADS_RESET
RESV1 31 AGND
AGND 20 AVSS
AGND 53 AVSS1
AGND 33 DGND

R101 1M RLD_OUT RLD_INV
C111 1.5nF RLD_INV RLD_OUT
R102 330k RLD_ELEC RLD_OUT

ISO7721DWVR (DNP)
U5
+3V3 1 VCC1
UART_RX 2 OUTA
UART_TX 3 INB
AGND 4 GND1
VCC2 8 VCC_ISO
INA 7 ISO_RX
OUTB 6 ISO_TX
GND2 5 GND_ISO

ISO UART (DNP)
J2
ISO_RX 1
ISO_TX 3
GND_ISO 2
VCC_ISO 4

C701 100nF (DNP) +3V3 AGND
C702 100nF (DNP) VCC_ISO GND_ISO
#FLG05 PWR_FLAG VCC_ISO
#FLG06 PWR_FLAG GND_ISO

SAFETY: with electrodes on the body and USB-C plugged in, run the laptop on battery,
unplugged from mains, or use a simulated input. Charge while worn only from
an isolated (Class II) charger. Keep the isolation gap under U5 free of copper.
LAYOUT: keep U1 and its supply away from the XIAO antenna.

ELECTRODES: one TRS per channel (T +, R -, S REF). All sleeves share RLD_ELEC,
so use 3 REF electrodes (9 total) or one (7 total). J6 FPC is in parallel with the jacks:
use one or the other. J6 1/5/9 + MP = AGND guards.

FLEX ARRAY (FPC 12P 0.5mm)
J6
AGND 1
ELEC1_P 2
ELEC1_N 3
RLD_ELEC 4
AGND 5
ELEC2_P 6
ELEC2_N 7
RLD_ELEC 8
AGND 9
ELEC3_P 10
ELEC3_N 11
RLD_ELEC 12
AGND MP

C112 10uF VREFP AGND
C113 100nF VREFP AGND
C114 22uF VCAP1 AGND
C115 1uF VCAP2 AGND
C116 1uF VCAP3 AGND
C117 1uF VCAP4 AGND

R103 10k +3V3 ADS_CS
R105 10k +3V3 ADS_RESET
R106 10k +3V3 ADS_PWDN

ADS1298: CH1-CH3 used, CH4-CH8 and TESTP tied to AVDD per datasheet (power down in firmware).
1, 2 or 3 channels by mode, no rework. HPF/LPF in firmware: capture ~0.5 Hz up at 4 kSPS+.
Lead-off detection and RLD channel selection are firmware settings.
Internal clock (CLKSEL high), internal 2.4 V reference.

Spec: emg_headband/hardware/emg_afe/README.md
1 to 3 channel surface EMG, ADS1298 + XIAO ESP32-S3, JLCPCB assembly

Sheet: /
File: emg_wearable_revA.kicad_sch

Title: EMG wearable AFE

Size: A3	Date: 2026-10-11	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/1